Skip-Enabled LMMSE-based Channel Estimation for Large-Scale Fluid Antenna-Enabled Cellular Networks

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***Abstract*—The concept of reconfigurable fluid antennas (FAs) is a potential and promising solution to enhance the spectral efficiency of wireless communication networks. Despite their many advantages, FA-enabled communications require a significant amount of spectral resources in order to select the most desirable position of the radiating element from a large number of prescribed locations. In this paper, we present an analytical framework for the outage performance of large-scale FA-enabled communications under limited coherence interval scenario. Under this framework, we propose a novel sequential linear minimum mean-squared error-based channel estimation method, that is performed for a limited number of FA ports, followed by data reception from the port with the strongest estimated channel. A complete analytical framework in terms of the outage probability is developed by using tools from stochastic geometry. Our results reveal that the proposed technique can provide significant performance gains, especially for FAs with a large number of ports.**

***Index Terms*—Fluid antenna, outage probability, LMMSE, port selection, stochastic geometry.**

I. Introduction

The ever-increasing demands of sixth generation (6G) wireless communication systems have spurred recent research activities on the concept of fluid antennas (FAs) [1]. Specifically, FAs consist of liquid radiating elements (e.g., Mercury, EGaIn, etc.), which are contained in a dielectric holder and can thus flow in different locations (i.e., a set of predefined ports) within its topological boundaries. Thus, FAs are capable to reversibly re-configure their physical configuration (i.e., size, shape and feeding) as well as their electrical properties (i.e., resonant frequency, bandwidth), providing a new degree of freedom in the design of wireless communication systems [1].

Throughout the literature, the potential benefits of the reconfigurable fluid-based antennas have been recently acknowledged, and the concept of FAs from an information/communication theoretic standpoint has been investigated. Particularly, the concept of FA is examined for point-to-point communication systems, where the performance in terms of outage probability [2] and ergodic capacity [3] is evaluated under spatially correlated channels. The authors show that a single FA with a large number of ports of half-wavelength or less separation between them can achieve performance similar to the conventional multi-antenna maximum ratio combining (MRC) system. These works are further extended for multi-user communications in [4], where the outage performance, the achievable rate, and the multiplexing gain of the considered topology are assessed; it is revealed that the overall network performance improves as the number of ports is increased at each receiver.

Nevertheless, the above studies assume that the FA-enabled communications have perfect knowledge of the channel state information (CSI) for all FA-associated links. In practice, such CSI needs to be acquired in each channel coherence interval at the cost of a channel training overhead that escalates with the number of FA ports [5]. Hence, channel estimation in FA-enabled networks is challenging but can also be impractical in some cases, due to the limited resources of FAs. Towards this direction, the problem of port selection for FA-enabled communications is investigated in [6], where several port selection algorithms are proposed for the case where the transmitter observes only a limited number of ports to reduce complexity. In addition, for a limited coherence interval scenario, channel estimation triggers a non-trivial trade-off between channel training duration and overall network performance. The impact of channel estimation on the network performance is investigated for different communication scenarios [7]–[10]. However, the deterministic configuration adopted by the above-mentioned works does not captures the irregularity associated with the actual deployments of cellular networks. Over the past decade, stochastic geometry (SG) has emerged as a powerful mathematical tool, which captures the random nature of large-scale networks and permits the analytical characterization of numerous performance metrics [11]. The impact of channel estimation in point-to-point single-input single-output ad-hoc systems [12] and random networks [13] is studied, by using linear minimum mean square error (LMMSE) channel estimation. In the context of FA systems, in [14], the authors assess the effect of FA technology on large-scale cellular networks, and study the trade-off imposed by the channel estimation on the outage performance; it is unveiled that an optimal number of FAs' ports maximizes the network performance with respect to the LMMSE-based channel estimation process.

Although few works exist on FA-enabled communications, to the best of our knowledge, none of these take into account the impact of channel estimation on the network performance from a large-scale point-of-view. Moreover, there is a lack of sophisticated techniques that efficiently utilize the limited spectral resources of FA-enabled communications.

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These limitations motivate the work presented in this paper. Initially, we shed light on the modeling and analysis of FA-enabled communications from a macroscopic point-of-view, by developing a rigorous mathematical framework based on SG. Moreover, we propose a novel skipped-enabled LMMSE-based channel estimation (SeCE) technique that leads to a channel estimation process of only a subset of "selected" ports from each FA, opposed to the technique developed in [14] that require channel estimation for all ports, followed by the data reception from the port with the strongest estimated channel. The developed channel estimation technique further reduces the required channel estimation coefficients, compared to the technique developed in [14], resulting in a significantly mitigated signaling overhead among the BSs and the UEs. Analytical expressions for the outage performance of FA-based UEs are derived and the performance of the considered system is investigated under different network parameters. Numerical results unveil that, compared to the conventional schemes, where channel estimation is performed for all FAs' ports, the proposed scheme significantly enhances the achieved network performance, especially for FAs with a large number of pre-defined ports.

Notation: Vectors are represented by boldface letters ($\boldsymbol{\alpha}$), and their conjugate transpose vectors are denoted by $\boldsymbol{\alpha}^\dagger$. The scenario where X is distributed as Y is denoted as $X \overset{d}{\sim} Y$. $J_0(\cdot)$ is the zero-order Bessel function of the first kind.

II. System Model

A. Network topology

Consider a single-tier cellular network with multiple randomly deployed BSs, modeled as points of a homogeneous Poisson point process (PPP), denoted as $\Phi = \left\{x_i \in \mathbb{R}^2, i \in \mathbb{N}^+\right\}$ with spatial density λ_b BS/m^2. Moreover, the locations of the UEs follow an arbitrary independent point process Ψ with spatial density $\lambda_u \gg \lambda_b$. All BSs transmit with power P (dBm) and are equipped with a single omnidirectional antenna, while all UEs are equipped with a single FA (see Section II-B). Orthogonal access like orthogonal time-division multiple access is assumed. Without loss of generality and by following Slivnyak's theorem [11], the analysis concerns the typical UE located at the origin but the results hold for all UEs of the network. We consider the nearest-BS association rule i.e., the typical UE communicates with its closest BS located at $x_0 \in \mathbb{R}^2$, referred as *tagged BS*, and its link with the typical UE is denoted as *typical link*.

B. Fluid antenna model

The adopted architecture of a fluid-based reconfigurable antenna is depicted in Fig. 1. Specifically, a drop of liquid metal is placed in a tube-like linear capillary filled with an electrolyte, within which the fluid is free to move. The location of the antenna (i.e., fluid metal) can be promptly switched to one of the N preset locations (also known as "ports"), that are evenly distributed along the linear dimension of a FA, $\kappa\lambda$, where λ is the wavelength of communication and κ is a scaling constant. As illustrated in Fig. 1, the distance between the first

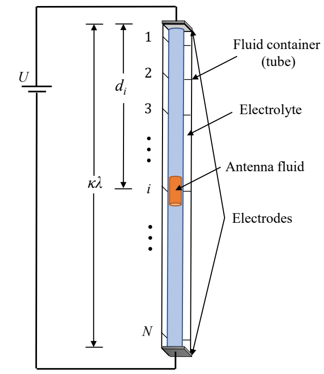

Fig. 1: A potential FA architecture.

port, that is considered as the reference port, and the i-th port can be measured as follows

$$d_i = \left(\frac{i-1}{N-1}\right)\kappa\lambda,\ \forall\ i \in \mathcal{N}, \tag{1}$$

where $\mathcal{N} = \{1, 2, \ldots, N\}$. Thus, the distance of the link between the i-th port and the tagged BS, is given by

$$r_i(\rho) = \sqrt{\rho^2 + \kappa^2\lambda^2\left(\frac{i-1}{N-1}\right)^2},\ \forall\ i \in \mathcal{N}, \tag{2}$$

where ρ is the distance of the typical link i.e., $\rho = \|x_0\|$, with probability density function (pdf) that is given by [11]

$$f(\rho) = 2\pi\lambda_b\rho\exp\left(-\pi\lambda_b\rho^2\right). \tag{3}$$

C. Channel model

The signals are assumed to experience both large-scale path-loss effects and small-scale fading. For the large-scale path-loss of the transmitted signals, we assume an unbounded singular path-loss model based on the distance r between a transmitter and a receiver i.e., $\ell(r) = r^a$, where $a > 2$ denotes the path-loss exponent. Moreover, a block fading channel mode is considered for the small-scale fading. In other words, the channel remains constant during a coherence time T_c, also known as *channel coherence interval*, and evolves independently from block to block. More specifically, the small-scale fading between the i-th port of the typical UE and its tagged BS, $|g_{0i}|$, is Rayleigh distributed. Due to the ability of FA's ports to be arbitrarily close to each other, the channels are assumed to be correlated. Hence, the channels observed by the N ports of the typical UE are given by

$$g_{0i} = \begin{cases} \sigma\alpha_1 + j\sigma\beta_1 & \text{if } i = 1, \\ \sigma\left(\sqrt{1-\mu_i^2}\alpha_i + \mu_i\alpha_0\right) + j\sigma\left(\sqrt{1-\mu_i^2}\beta_i + \mu_i\beta_0\right) & \text{otherwise}, \end{cases} \tag{4}$$

where $i \in \mathcal{N}$, $\alpha_N, \beta_1, \ldots, \beta_N$ are all independent Gaussian random variables with zero mean and variance of $\frac{1}{2}$; μ_i is the autocorrelation parameter that can be selected appropriately to determine the channel correlation between the i-th port and the reference (i.e., first) port. In this work, the autocorrelation parameter is given by [2]

$$\mu_i = \begin{cases} 0 & \text{if } i = 1, \\ J_0\left(\frac{2\pi(i-1)}{N-1}\kappa\right) & \text{otherwise}. \end{cases} \tag{5}$$

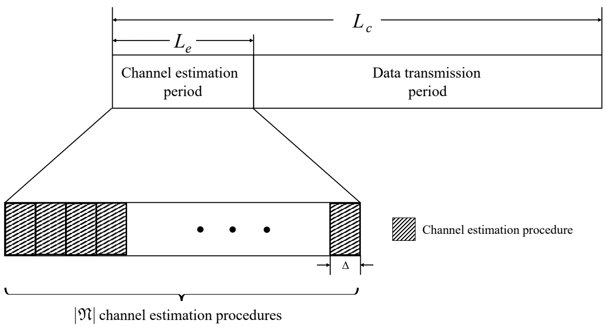

Fig. 2: Representation of a block fading channel consisting of channel estimation and data transmission periods.

D. Skip-enabled LMMSE-based channel estimation scheme

For the considered limited coherence interval scenario, a novel LMMSE-based channel estimation technique is developed, namely SeCE technique. In contrast to conventional approaches, where the channel estimation for all FAs' ports is performed, the proposed SeCE technique requires a channel estimation process for only a subset of "selected" ports. In particular, the serving BS initially broadcasts pilot-training symbols to the reference (i.e., first) port, estimating their link's channel coefficients by employing a LMMSE channel estimation technique. Then, by leveraging the strong spatial correlation between adjacent FA's ports, the channels for the subsequent ν ports are assumed to be equal to the estimated channel of the previously "selected" port, thereby the channel estimation process related to these ports is skipped. Hence, the SeCE technique proceeds with the channel estimation process of the link between the serving BS and the $(\nu+2)$-th port. The previous process is repeated until channel estimation for all links between the serving BS and the FA's ports is obtained. The set defined by the ports for which the serving BS performs LMMSE channel estimation, represents the set of "selected" FA's ports i.e., $\mathfrak{N} \subseteq \mathcal{N}$. Therefore, our proposed technique requires channel estimation for only $|\mathfrak{N}| = \lceil \frac{N}{\nu+1} \rceil$ links, opposed to the conventional techniques that require channel estimation for N links. Despite the alleviated signaling overhead achieved with the SeCE technique, the reduced channel resolution caused by the sparse channel estimation process, jeopardizes the observed SINR. Hence, the employment of our proposed SeCE technique triggers a non-trivial trade-off between mitigating the signaling overhead related to the channel estimation process and reducing the observed SINR. Note that, by considering $\nu = 0$, our proposed channel estimation scheme becomes the conventional LMMSE channel estimation scheme, where channel estimation is performed for all FAs' ports, and referred as *Skip-less CE technique*.

III. FA-enabled Communications Under SeCE Technique

In this section, we evaluate the outage performance of FA-enabled communications in the context of the proposed SeCE scheme for a scenario with a limited coherence interval. According to the adopted block fading model, the length of each coherence interval/block (L_c channel uses) is divided into two sub-blocks for channel estimation and data communication, as illustrated in Fig. 2. In particular, we consider L_e channel uses per block for pilot-training symbols to enable channel estimation, and $L_t = L_c - L_e$ channel uses are used for downlink data transmission. In the following sections, we elaborate in detail the channel estimation and data transmission periods in the context of the proposed SeCE technique. Finally, analytical expressions for the achieved outage probability are derived, by leveraging tools from SG.

A. Channel estimation and Data Transmission period

1) Channel Estimation period: Under the proposed SeCE technique, the entire channel estimation period, L_e, is divided into $|\mathfrak{N}|$ symmetric segments of $\Delta = \frac{L_e}{|\mathfrak{N}|}$ consecutive symbols (see Fig. 2). During each segment, the channel between a single FA port and the serving BS is estimated[1]. Therefore, the baseband equivalent received pilot signal at the i-th port of the typical UE, is given by

$$\mathbf{y}_i = \sqrt{\frac{\Delta P}{\ell(r_i(\rho))}} g_{0i} \mathbf{X}_0 + \sum_{\substack{j\in\mathbb{N}^+ \\ x_j \in \Phi \setminus x_0}} \sqrt{\frac{P}{\ell(\|x_j\|)}} g_{ji} \mathbf{X}_j + \eta_0, \quad (6)$$

where $\mathbf{X}_0$ is a deterministic $\Delta \times 1$ training symbol vector, $\mathbf{X}_j \overset{d}{\sim} \mathcal{CN}(\mathbf{0}_{\Delta\times 1}, \mathbf{I}_\Delta)$ is a transmission symbol vector from node j with channel $g_{ji} \overset{d}{\sim} \mathcal{CN}(0,1)$, and $\eta_0 \overset{d}{\sim} \mathcal{CN}(\mathbf{0}_{\Delta\times 1}, N_0 \mathbf{I}_\Delta)$ is the additive white Gaussian noise (AWGN) vector. Then, under the low-complexity LMMSE estimator, which is optimal among the class of linear estimators, the estimate of g_{0i} conditioned on ρ, is given by

$$\widehat{g}_{0i}|_\rho = \frac{\sqrt{\frac{\Delta P}{\ell(r_i(\rho))}}\sigma^2}{\frac{\Delta P}{\ell(r_i(\rho))}\sigma^2 + N_0 + P\mathbb{E}(\mathcal{I})} y_i, \quad (7)$$

where $\mathcal{I}$ is the instantaneous power of the interference at the typical UE, and is given by $\mathcal{I} = \sum_{\substack{j\in\mathbb{N}^+ \\ x_j\in\Phi\setminus x_0}} \frac{1}{\ell(\|x_j\|)} |g_{ji}|^2$. According to the Campbell's Theorem [11] and conditioning on ρ, $\mathbb{E}(\mathcal{I})$ can be evaluated as

$$\mathbb{E}(\mathcal{I}) = 2\pi\lambda_b \mathbb{E}\left[|g_{ji}|^2\right] \int_{r_i(\rho)}^{\infty} r^{1-a} \mathrm{d}r = 2\pi\lambda_b\sigma^2 \frac{r_i(\rho)^{2-a}}{a-2}, \quad (8)$$

where $r_i(\rho)$ is given by (2). The channel estimation error can then be derived as $e_i|_\rho = g_{0i} - \widehat{g}_{0i}|_\rho$, where $e_i|_\rho \overset{d}{\sim} \mathcal{CN}\left(0, \sigma_e^2|_\rho\right)$, and $\widehat{g}_{0i}|_\rho$ and $e_i|_\rho$ are uncorrelated [8]. The variance of the channel estimation conditioned on ρ, $\sigma_e^2|_\rho = \mathbb{E}\left[\left(g_i - \hat{g}_i|_\rho\right)^2\right]$, is evaluated in the following proposition.

Proposition 1. *By employing the SeCE technique, the variance of the channel estimation error observed at the i-th port of the typical UE, conditioned on ρ, is given by*

$$\sigma_e^2|_\rho = \left(1 + \frac{\Delta}{\frac{r_i^a(\rho)}{\epsilon} + 2\pi\lambda_b \frac{r_i^2(\rho)}{a-2}}\right)^{-1}. \quad (9)$$

Proof. The proof of the above Proposition can be found in [15, Proposition 1]. □

[1]More sophisticated vector-based channel estimation schemes can be used but this estimation process is sufficient for the purpose of this work.

Remark 1. *The variance of channel estimation error is a non-negative increasing concave function with respect to the number of ports N. For the extreme case where $N \to \infty$, the variance of the channel estimation error (i.e., channel estimation quality) becomes equal to one i.e., $\sigma_e^2|_\rho \approxeq 1$.*

2) Data transmission period: Regarding the data transmission period (i.e., for the rest of the coherence block after the channel estimation process), all BSs communicate with their associated UEs. By aiming to achieve the best reception performance, we assume that the FA's location is switched to the port that is estimated to provide the strongest channel. Hence, the FA's location of the typical UE is switched to the port that satisfies

$$\mathtt{i} = \arg\max_{i \in \mathfrak{N}} \left\{ \left| \widehat{g}_{0i} \right| \right\}. \tag{10}$$

For the sake of simplicity, we consider that both the data and the pilot-training symbols are transmitted with the same power P (dBm). Thus, the received signal at the $\mathtt{i}$-th port of the typical UE during the n-th channel use, is given by

$$d_{\mathtt{i}}[n] = \sqrt{\frac{1}{r_{\mathtt{i}}^a(\rho)}} \widehat{g}_{0\mathtt{i}} s_0[n] + \sqrt{\frac{1}{r_{\mathtt{i}}^a(\rho)}} e_{\mathtt{i}} s_0[n] + \sum_{\substack{j \in \mathbb{N}^+ \\ x_j \in \Phi \setminus x_0}} \sqrt{\frac{1}{\|x_j\|^a}} g_{j\mathtt{i}} s_j[n] + \eta_0[n], \tag{11}$$

where $s_0[n]$ and $s_j[n]$ represent independent Gaussian distributed data symbols from the tagged and the j-th interfering BS, respectively, and $n \in \{L_e + 1, \ldots, L_c\}$. Note that, the first term of (11) is known at the receiver, while the remaining terms are unknown and are treated as noise. Thus, an estimate of $s_0[n]$ can be formulated as $\hat{s}_0[n] = \sqrt{r_{\mathtt{i}}^a(\rho)} \frac{(\widehat{g}_{0\mathtt{i}})^*}{\left|\widehat{g}_{0\mathtt{i}}\right|^2} d_{\mathtt{i}}[n]$, from which the SINR observed at the $\mathtt{i}$-th port can be written as

$$\mathrm{SINR}_{\mathtt{i}} = \frac{\frac{\epsilon}{r_{\mathtt{i}}^a(\rho)} \left|\widehat{g}_{0\mathtt{i}}\right|^2}{\sum_{\substack{j \in \mathbb{N}^+ \\ x_j \in \Phi \setminus x_0}} \frac{\epsilon}{\|x_j\|^a} \left|g_{ji}\right|^2 + \frac{\epsilon}{r_{\mathtt{i}}^a(\rho)} \sigma_e^2|_\rho + 1}, \tag{12}$$

where $\epsilon = \frac{\sigma^2 P}{N_0}$ is the transmit signal-to-noise ratio (SNR).

Remark 2. *For the case of $N \to \infty$, the inadequate channel estimation quality (see Remark 1) induces a substantial reduction of the SINR observed by the FA-based UEs (see* (12)*), compromising the network performance.*

Based on Remark 1 and Remark 2, although the increased number of FA ports initially enhances the receive diversity gain and thereby the network performance, beyond a critical point $N^* \in \mathcal{N}$, a further increase of the number of FA ports leads to an attenuated channel estimation quality, jeopardizing the overall network performance. Hence, the number of FA ports triggers a trade-off between improving the network's outage performance and reducing the channel estimation quality.

B. Channel and Interference Statistical Properties

We start with the statistical properties of the estimated channels under the SeCE technique. In particular, the joint pdf and cumulative distribution function (cdf) of $|\hat{g}_{01}|, \ldots, \left|\hat{g}_{0|\mathfrak{N}|}\right|$, conditioned on ρ, are given in the following lemma.

Lemma 1. *The conditional joint cdf and pdf of $|\widehat{g}_{01}|, \ldots, \left|\widehat{g}_{0|\mathfrak{N}|}\right|$ for $i \in \mathfrak{N}$, are given by*

$$F_{|\widehat{g}_{01}|, \ldots, |\widehat{g}_{0|\mathfrak{N}|}|}(\tau_1, \ldots, \tau_{|\mathfrak{N}|} | \rho) = \int_0^{\frac{\tau_1^2}{\widetilde{\sigma}_1^2}} \exp(-t) \prod_{j \in \mathfrak{N}} \left[1 - Q_1\left(\sqrt{2\mu_j^2 \frac{\widetilde{\sigma}_1^2}{\widetilde{\sigma}_j^2} t}, \sqrt{\frac{2}{\widetilde{\sigma}_j^2}} \tau_j\right)\right] \mathrm{d}t \tag{13}$$

and

$$f_{|\widehat{g}_{01}|, \ldots, |\widehat{g}_{0|\mathfrak{N}|}|}(\tau_1, \ldots, \tau_{|\mathfrak{N}|} | \rho) = \prod_{\substack{i \in \mathfrak{N} \\ (\mu_1 \triangleq 0)}} \frac{2\tau_i}{\widetilde{\sigma}_i^2} \exp\left(-\frac{\tau_i^2 + \mu_i^2 \tau_1^2}{\widetilde{\sigma}_i^2}\right) J_0\left(\frac{2\mu_i \tau_1 \tau_i}{\widetilde{\sigma}_i^2}\right), \tag{14}$$

respectively, where $\tau_1, \ldots, \tau_{|\mathfrak{N}|} \geq 0$, $Q_1(\cdot, \cdot)$ is the first-order Marcum Q-function, and $\widetilde{\sigma}_i^2 = \sigma^2(1 - \mu_i^2) + \sigma_e^2|_\rho$.

Proof. See Appendix A. ☐

Multi-user interference has been shown to be the main limiting factor in large-scale networks [11]. Although the performance of a communication network can be easily evaluated in the case of PPP with independent fading channels, in most relevant (realistic) models, it is either impossible to analytically analyze or cumbersome to evaluate numerically. Hence, for analytical tractability purposes, in this paper, we assume that the multi-user interference of large-scale wireless networks is approximated by its mean value i.e., $\mathcal{I} = \mathbb{E}[\mathcal{I}]$.

C. Outage performance

The downlink outage performance $\mathcal{P}_o(R)$ of a network can be described as the probability that the maximum mutual information of the channel between the typical UE and its serving BS is smaller than a target rate R (data bits/channel use). This performance can be mathematically described with the probability $\mathbb{P}\left[\left(1 - \frac{L_t}{L_c}\right) \log\left(1 + \mathrm{SINR}\right) < R\right]$, where $\left(1 - \frac{L_t}{L_c}\right)$ is the fractional amount of time (relative to the total frame length) used for data transmission. Hence, $\mathcal{P}_o(R)$ can be expressed as follows

$$\mathcal{P}_o(R) = \mathbb{E}_\rho\left[\mathbb{P}\left[|\widehat{g}_{0\mathtt{i}}| < \sqrt{\vartheta r_{\mathtt{i}}^a(\rho)\left(\mathbb{E}[\mathcal{I}] + \frac{\sigma_e^2|_\rho}{r_{\mathtt{i}}^a(\rho)} + \frac{1}{\epsilon}\right)} \middle| \rho\right]\right] = \mathbb{E}_\rho\left[\mathbb{P}\left[|\widehat{g}_{0\mathtt{i}}| < \sqrt{\Theta_{\mathtt{i}}} \middle| \rho\right]\right], \tag{15}$$

where $\vartheta = 2^{\frac{R}{1 - \frac{L_t}{L_c}}} - 1$ and $\Theta_i = \vartheta r_i^a(\rho)\left(\mathbb{E}[\mathcal{I}] + \frac{\sigma_e^2|_\rho}{r_i^a(\rho)} + \frac{1}{\epsilon}\right)$. In the following lemma, an analytical expression for the conditional outage performance i.e., $\mathcal{P}_o(R|\rho) = \mathbb{P}\left[|\widehat{g}_{0\mathtt{i}}| < \sqrt{\Theta_{\mathtt{i}}} | \rho\right]$, of the considered system model is derived.

Lemma 2. *The conditional outage probability when utilizing the proposed SeCE technique, is given by*

$$\mathcal{P}_o(R|\rho) = \int_0^{\frac{\Theta_1^2}{\widetilde{\sigma}_1^2}} \exp(-t) \prod_{j \in \mathfrak{N}} \left[1 - Q_1\left(\sqrt{2\mu_j^2 \frac{\widetilde{\sigma}_1^2}{\widetilde{\sigma}_j^2} t}, \sqrt{\frac{2}{\widetilde{\sigma}_j^2}} \Theta_j\right)\right] \mathrm{d}t, \tag{16}$$

where $\Theta_j = \vartheta r_j^a(\rho)\left(\mathbb{E}[\mathcal{I}] + \frac{\sigma_e^2|_\rho}{r_j^a(\rho)} + \frac{1}{\nu}\right)$; $r_j(\rho)$ and $\sigma_e^2|_\rho$ depict the length of the typical link and the variance of the channel estimation, that are given by (2) *and* (9)*, respectively, and $\widetilde{\sigma}_j^2 = \sigma^2(1 - \mu_j^2) + \sigma_e^2|_\rho$.*

Proof. The conditional outage probability expression can be calculated as

$$\mathcal{P}_o(R|\rho) = \mathbb{P}\left[|\widehat{g}_{0\mathtt{i}}| < \sqrt{\Theta_{\mathtt{i}}}\middle|\rho\right]$$
$$= \mathbb{P}\left[|\widehat{g}_{01}| < \sqrt{\Theta_1}, \dots, |\widehat{g}_{0|\mathfrak{N}|}| < \sqrt{\Theta_{|\mathfrak{N}|}}|\rho\right].$$

Thus, by substituting $\tau_1 = \sqrt{\Theta_1}, \dots, \tau_N = \sqrt{\Theta_N}$ into the joint cdf that is given in Lemma 1, which completes the proof. □

In the following Theorem, we evaluate the outage performance of the considered system model, by un-conditioning the expression in Lemma 2 with the pdf of the distance from the typical UE to its serving BS.

Theorem 1. *The outage probability achieved by the proposed SeCE technique in cellular networks with FA-based UEs, is given by*

$$\mathcal{P}_o(R) = \prod_{i\in\mathfrak{N}} \int_0^\infty \mathcal{P}_o(R|\rho) 2\pi\lambda_b\rho \exp\left(-\pi\lambda_b\rho^2\right) \mathrm{d}\rho,$$

where $\mathcal{P}_o(R|\rho)$ represents the achieved outage probability conditioned on the distance of the typical UE from its serving BS ρ (2), that is given in Lemma 2.

Proof. The expression for the outage probability can be obtained by firstly substituting the expression derived in Lemma 2 into (15). Then, the final expression can be derived by getting rid of the condition on ρ by utilizing (3). □

IV. Numerical Results

In this section, we provide numerical results to verify our model and illustrate the performance of large-scale FA-enabled communications. Specifically, we consider the following parameters: $\lambda_b = 15$ BS/km^2, $\lambda_u = 30$ UE/km^2, $a = 4$, $\lambda = 6$ cm, $\sigma^2 = -60$ dB, and $L_c = 2500$ channel uses.

Fig. 3 plots the outage probability $P_o(R)$, in terms of the transmit power P, for different number of FAs' ports $N = \{5, 15, 25\}$ and scaling constants $\kappa = \{0.2, 0.5\}$. It is clear from the figure that the outage probability decrease with κ, since a larger FA architecture i.e., larger distance between the FA ports, leads to the limitation of the negative effect of the spatial correlation between the ports' channels on the network performance. Furthermore, the performance increases with the number of FAs' ports since a larger value of N implies higher receive diversity gain, thereby UEs observe an enhanced SINR. Also, we can observe that the outage probability asymptotically converges to a constant value. This is due to the fact that as the transmission power of the nodes increases, the additive noise in the network becomes negligible. Moreover, Fig. 3 shows the impact of actual interference on the network performance. The small deviation between these curves indicates that, the adoption of the simplified interference model provides lower complexity methodology for evaluating the system performance, without being significantly deficient in accuracy. Finally, the agreement between the theoretical curves (solid and dotted lines) and the simulation results (markers) validates our analysis.

Fig. 4 demonstrates the outage probability versus the number of FAs' ports, N, for different $\nu = \{1, 2, 3\}$. We can easily observe that for small number of FAs' ports, the outage performance reduces with N, while beyond a critical point N^*, the outage performance increases. This was expected since the allocated number of training-pilot symbols for the channel estimation of each port is decreases with N, and thus, the quality of the channel estimation is decreased (i.e., $\sigma^2_{e|\rho} \to 1$), diminishing the achieved network performance. Fig. 4 also highlights the effect of the proposed SeCE technique on the achieved outage performance. An important observation is that the critical number of ports, N^*, increases with ν, since a larger number of ports for which a UE can skip their channel estimation processes, leads to the successful channel estimation process of FAs with a larger number of ports. Furthermore, beyond the case where $1 \leq N \leq 7$, the SeCE technique leads to the improvement of the network performance compared to the conventional Skip-less CE technique (i.e., $\nu = 0$). This was expected since, a larger value of ν triggers the allocation of more time for pilot-based training for the remaining ports, enhancing the outage performance. Specifically, the employment of the SeCE technique with $\nu = 3$ enhances the network performance by 58.52%, compared with the conventional technique for the scenario where $N = 40$. For comparison purposes, we also present the outage performance obtained with a perfect (a-priori) CSI [4], denoted as "Perfect CSI". We can easily observe that, in contrast to the scenario considered with channel estimation error, the network performance is constantly increases with N, since the negative effect of channel estimation quality on the network's performance is neglected.

Fig. 5 evaluates the outage performance with respect to the density of BSs for different values of wavelength constants $\lambda = \{3, 6\}$ cm. As mentioned before, the performance experienced by a FA-based UE can be improved by adopting larger FA architectures i.e., a larger λ. Another interesting observation is that the outage performance initially decreases with λ_b but, after a certain value of λ_b, it starts to increase. This observation is based on the fact that at low density values, the increased number of BSs leads to a reduction of the distance between the UEs and their serving BSs, thereby the observed SINR at the UEs is enhanced. On the other hand, for high BSs' densities, the interference caused by the active UEs increases, thereby reducing the ability of the BSs to decode the received signal.

V. Conclusion

In this paper, we presented an analytical framework based on SG to study the outage performance of FA-based UEs in the context of large-scale homogeneous cellular networks. Aiming at improving the channel estimation quality and thereby enhancing the network's performance in practical limited coherence interval scenarios, we proposed a novel LMMSE-based channel estimation technique, where a channel estimation process of only a small number of FA ports is performed. Analytical expressions for the outage performance in the context of our proposed technique were derived, and the impact of nodes density, block length, and number of FAs'

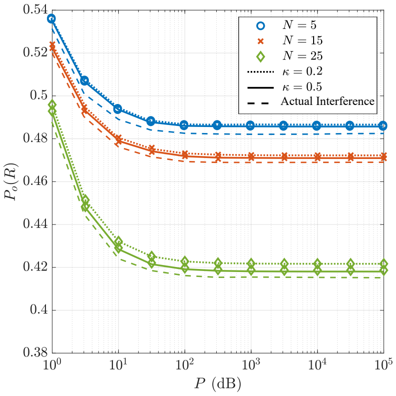

Fig. 3: Outage performance versus the transmit power P (dB) for different N and κ; $\nu = 1$.

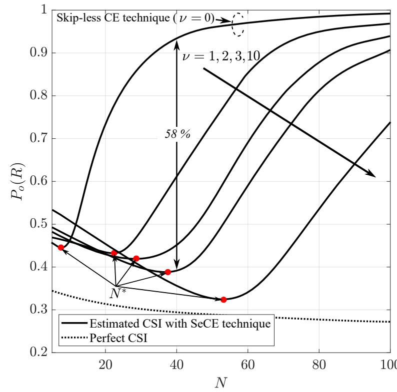

Fig. 4: Outage performance versus the number of ports N for different ν.

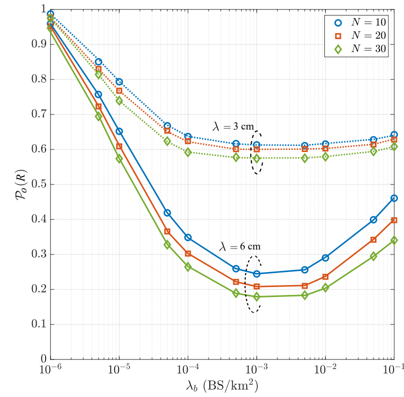

Fig. 5: Outage performance versus the BSs density λ_b (BS/km^2) for different N and λ; $\kappa = 1$.

ports has been discussed. Our results reveal that the proposed scheme provides significant performance gains compared to the conventional case, whilst keeping the complexity low.

Appendix A
Proof of Lemma 1

Under the proposed LMMSE-based SeCE technique, the channel experienced by the i-th port of the typical UE can be expressed as $\widehat{g}_{0i}|_\rho = g_{0i} + e_i|_\rho$, where $g_{0i} \overset{d}{\sim} \mathcal{CN}(0,\sigma^2)$ and $e_i|_\rho \overset{d}{\sim} \mathcal{CN}\left(0,\sigma_e^2|_\rho\right)$, and hence, $\widehat{g}_{0i}|_\rho \overset{d}{\sim} \mathcal{CN}\left(0,\widetilde{\sigma}_i^2\right)$, with $\widetilde{\sigma}_i^2 = \sigma^2(1-\mu_i^2) + \sigma_e^2|_\rho$. Then, the amplitude of the estimated channels, $|\widehat{g}_{0i}|$, is Rayleigh distributed, with pdf $f_{|\widehat{g}_{0i}|}(\tau) = \frac{2\tau}{\widetilde{\sigma}_i^2}\exp\left(-\frac{\tau^2}{\widetilde{\sigma}_i^2}\right)$, where $\mathbb{E}\left[|\widehat{g}_{0i}|^2\right] = \widetilde{\sigma}_i$. As it already mentioned, the channels $\{\widehat{g}_{0i}\}$ are correlated due to the capability of FA's ports to be arbitrarily close to each other. In particular, the amplitude of the estimated channel $|\widehat{g}_{02}|$, conditioned on ρ, x_0 and y_0, follows a Rice distribution, i.e.,

$$\begin{aligned} &f_{|\widehat{g}_{02}||\{x_0,y_0\}}(\tau_2|\rho,x_0,y_0) \\ &=\frac{2\tau_2}{\widetilde{\sigma}_i^2}\exp\!\left(-\frac{\tau_2^2+\mu_2^2(x_0^2+y_0^2)}{\widetilde{\sigma}_i^2}\right)J_0\!\left(\frac{2\mu_2\sqrt{x_0^2+y_0^2}\,\tau_2}{\widetilde{\sigma}_i^2}\right), \quad (17) \end{aligned}$$

where $\tau_2 \geq 0$. By substituting $\tau_1 \to \sqrt{x_0^2+y_0^2}$ and since $x_0, y_0, |\widehat{g}_{02}|, \dots, |\widehat{g}_{0|\mathfrak{N}|}|$ are all independent between each other, the joint pdf of the estimated channels, conditioned on $|\widehat{g}_{01}|$, can be expressed as

$$\begin{aligned} &f_{|\widehat{g}_{02}|,\dots,|\widehat{g}_{0|\mathfrak{N}|}|||\widehat{g}_{01}|}(\tau_2,\dots,\tau_{|\mathfrak{N}|}|\rho,\tau_1) \\ &\quad= \prod_{i\in\mathfrak{N}} \frac{2\tau_i}{\widetilde{\sigma}_i^2}\exp\left(-\frac{\tau_i^2+\mu_2^i\tau_1^2}{\widetilde{\sigma}_i^2}\right) I_0\left(\frac{2\mu_i\tau_1\tau_i}{\widetilde{\sigma}_i^2}\right). \quad (18) \end{aligned}$$

Then, the final expression can be achieved by unconditioning the above expression with the pdf of $|\widehat{g}_{01}|$ i.e., $f_{|\widehat{g}_{02}|,\dots,|\widehat{g}_{0|\mathfrak{N}|}|||\widehat{g}_{01}|}(\tau_2,\dots,\tau_{|\mathfrak{N}|}|\rho,\tau_1)f_{|\widehat{g}_{01}|}(\tau_1)$, which gives the desired expression. Regarding the joint cdf of $|\widehat{g}_{0i}|$, based on the definition, this is given by

$$\begin{aligned} &F_{|\widehat{g}_{01}|,\dots,|\widehat{g}_{0|\mathfrak{N}|}|}(\tau_1,\dots,\tau_{|\mathfrak{N}|}|\rho) \\ &= \int_0^{\tau_1}\cdots\int_0^{\tau_{|\mathfrak{N}|}} f_{|\widehat{g}_{01}|,\dots,|\widehat{g}_{0|\mathfrak{N}|}|}(\tau_1,\dots,\tau_{|\mathfrak{N}|}|\rho)\mathrm{d}\tau_1\cdots\mathrm{d}\tau_{|\mathfrak{N}|}. \end{aligned}$$

By using the derived expression for the joint pdf and by using the transformation $t = \frac{\tau_1^2}{\widetilde{\sigma}_1^2}$, the final expression can be derived.

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